

Nakul Rajpal

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Education

Northeastern University, Khourey College of Computer Sciences, Boston, MA September 2023 - May 2027
Candidate for Bachelor of Science in Computer Science

- **Relevant Coursework:** Foundations of Software Engineering, Artificial Intelligence in Human-Computer Interaction, Distributed Systems, Algorithms and Data, Object Oriented Design, Fundamentals of Data Science, Computer Systems

Skills

Languages: Python, Java, C#, C, C++, R, JavaScript, SQL

Tools: Figma, React, Android Studio, Pytorch, Tensorflow, Arduino, CI/CD, RestAPI

Work Experience

Undergraduate Researcher, *Civic AI Lab* – Boston, MA April 2024 - Present

- Led 20+ semi-structured interviews investigating gig worker upskilling strategies within the AI economy; co-authored research paper currently **Under Review** for CHIWORK 2026.
- Architecting **UserWeave**, a five-stage co-design system for web development that leverages LLM-driven synthesis to facilitate collaborative decision-making. (**In Preparation**, UIST 2026).

Undergraduate Research Intern, *North Dakota State University* – Fargo, ND June 2025 - September 2025

- Developed an anomaly detection pipeline combining CTGAN-based tabular data synthesis and deep autoencoder models to detect IoT-based malicious traffic.
- Used synthetic data to train anomaly detection models achieving over 95% accuracy, employing MAD-based thresholding and extensive evaluation on 6M+ test samples.

Undergraduate Research Intern, *University of Nevada* - Las Vegas, NV June 2024 - August 2024

- Constructed Federated Learning frameworks to optimize power generation predictions from wind turbine data, improving sustainability efforts.
- Designed and trained four models using an Artificial Neural Network, reducing training time and increasing prediction accuracy.

Projects

BillTracks | *Supabase, Docker, Python, OpenAI API, RAG, MongoDB, TypeScript* October 2024 - Present

- Built a full-stack web app that simplifies congressional bills using AI-powered summaries (OpenAI API + RAG).
- Enabled real-time bill tracking with Supabase and MongoDB, providing dynamic updates on legislative progress.
- Deployed using Docker for scalable infrastructure and optimized performance, averaging 40,000 monthly visits.
- Website: Bill Tracks

Dyslexia Translator | *Typescript, JavaScript, Prompt Engineering, LLM's* Oct 2024 - December 2024

- Developed a web-based interface to translate homework problems into dyslexia- and dyscalculia-friendly formats, promoting educational accessibility.
- Utilized Large Language Models (LLMs) and prompt engineering to adapt language and numerical content for improved comprehension. Built with TypeScript and JavaScript, with a focus on user-centric design and accessibility best practices.

Skillshift | *Python, Typescript, React, Prompt Engineering, LLM's* Oct 2024 - January 2025

- Developed a Python-based app that uses Llama API and LLMs to help gig workers upskill and transition into new industries. Applied prompt engineering to generate personalized learning paths and career recommendations.
- Utilized TypeScript and React for a responsive and scalable user experience.